

Problem 10.27

A fund earns interest at a force of interest of

$$\delta_t = kt.$$

Ryan invests \$1,000 at time $t = 0$. After 8 years, Ryan has \$1,896.50. Calculate k .

Solution

For a time-varying force of interest, the accumulation factor from time 0 to time 8 is

$$\exp \left\{ \int_0^8 \delta_t dt \right\}.$$

Therefore,

$$1896.50 = 1000 \exp \left\{ \int_0^8 kt dt \right\}.$$

Evaluating the integral gives

$$\int_0^8 kt dt = \left. \frac{kt^2}{2} \right|_0^8 = \frac{64k}{2} = 32k.$$

Hence,

$$1896.50 = 1000e^{32k}.$$

Dividing by 1000 and taking natural logarithms,

$$\begin{aligned} 1.8965 &= e^{32k}, \\ \ln(1.8965) &= 32k. \end{aligned}$$

Therefore,

$$\begin{aligned} k &= \frac{\ln(1.8965)}{32} \\ &\approx 0.020000315067. \end{aligned}$$

Thus,

$$\boxed{k \approx 0.0200003 \approx 2\%}.$$